Net Deed Plotter Cheat Sheet

A quick reference for plotting deeds in Greenbriar Net Deed Plotter, built from the software manual. Get the software at deedplotter.com. To turn a legal description into these calls automatically, use the Net Deed Plotter Formatter at mccrackenlandservices.com.

1. Entering metes and bounds

Start at the first "thence" and enter one course per line. A course is quadrant, degrees, a period, then minutes and seconds as four digits, the second quadrant, a space, and the distance.

```
North 22 degrees 13 minutes 10 seconds East,
200.7 feet
N22.1310E 200.7  then press Enter
```

Minutes and seconds are always two digits each. Five minutes and three seconds is `0503`, so a call might read `S4.0503W 96.2`.

2. Varas and feet

Older deeds measure in varas. Put a **V** at the end of the distance so the plot figures acreage and scales right. Feet need no suffix.

```
North 22 degrees 13 minutes 10 seconds East,
200.7 varas
N22.1310E 200.7V
```

4. Units and directions it accepts

Distances: feet, varas, chains, rods, poles, perches, links, and meters. **Directions:** bearings, azimuths, deflections, and interior angles.

3. Entering a section (aliquot parts)

Type the aliquot parts, section, township, and range in abbreviated form, separated by commas with no spaces, then press Enter. Net Deed Plotter converts it to metes and bounds.

```
All of section 23, T10N, R13W
all,23,10N,13W  then press Enter

N/2 of the SE/4 of sec 9, T8N, R15W
n,se,9,8n,15w
```

ALL	All of the section
N / W	North half / West half
SW	Southwest quarter
N,S	North half of the south half
E,SW	East half of the southwest quarter
S,NE,SE,SW	S/2 of NE of SE of SW
N251,S800	North 251 ft of the south 800 ft

List the aliquot parts in the wording order (largest tract last). To **tie** a metes-and-bounds call to a section corner, begin with a `/` then corner, quarter-quarter, quarter, section, township, range, for example `/se,n,s,10,4n,5w`.

5. Curves

Type the word **Curve** on its own line, then double-click it (or press Ctrl+Enter) to open the Curve Data Entry Form. Press Tab to move between fields. You are not expected to have every field.

- **Curve direction**: R or L (or C for concave-to a quadrant). Required.
- **Radius**: in feet. If it is missing, enter `?` as a last resort, or enter a degree of curvature like `3.05d` and let the program compute the radius.
- **Arc length**: the distance around the curve. Never less than the chord.
- **Delta (central angle)**: the angle the curve sweeps, in degrees-minutes-seconds, such as `132.0732`.
- **Chord direction**: a bearing, such as `N50.2459W`. **Chord distance**: straight end to end, never more than the arc.
- **Back tangent, ahead tangent, radial direction**: bearings used for non-tangent curves.

> **Minimum curve data.** A normal (tangent) curve usually draws with the direction, the radius, and one more element (arc, delta, or chord distance). A non-tangent curve, or any curve that is the first call in a tract, must also include one bearing: chord, tangent, radial, or ahead-tangent direction.

6. Draw a tract with the mouse (Advanced)

Handy when a new tract begins at a point you can see on a tract already on the map, like the intersection of a road and a railroad or a shared corner, and the deed gives no written tie to a section.

- Have the first tract on screen so the starting point is visible.
- Open the **Advanced** menu, then **Draw Tract with Mouse**, and click your beginning point. To land exactly on a corner of an existing tract, hover until the cursor becomes a hand, then click.
- Click a few rough corners, close the loop by clicking the first point again, and answer Yes. The rough calls drop into the Deed Call Editor starting with `@0`.
- In the editor, delete the guessed calls so the cursor sits at your true point of beginning, then type the real calls.

7. The "?" for a missing call

If a deed is missing a call or a radius, put a `?` where that call would go and let Net Deed Plotter solve it. For a missing curve, follow the `?` with the curve direction and radius, no spaces.

```
?L5729.58   a curve to the left, radius 5729.58 ft
```

Only one `?` is allowed per tract.

> **Careful.** Solving with a `?` drives the closure error to near zero, so it can hide a real mistake. Confirm the solved value against the deed before you trust it.

8. Map and workflow tips

- Press `F2` to view the map.
- **Scale**: pick Scale in the directory tree. It is inches per feet, so 100 means 1 inch = 100 feet. Smaller number, larger map.
- **Color**: click inside the tract until it is outlined in green, then use Fill (solid, hatch, or checker) or Lines for color.
- **Label courses**: Map, then Deed Calls on Lines.
- **Bad closure**: the Analyze option helps find the faulty call. A bad call shows in red.
- Add any text notes **last**, after the map is scaled and placed.
- Net Deed Plotter reports area, net area, closure, and precision.

9. Minutes and seconds are base 60

A bearing is degrees, minutes, and seconds. There are 60 minutes in a degree and 60 seconds in a minute, so a fraction of a minute is seconds, not a decimal.

- Half a minute is **30 seconds**, not 0.5. "45 and a half minutes" is 45 minutes 30 seconds, written as the pair `4530`.
- A quarter minute is 15 seconds. Three quarters is 45 seconds.

```
North 12 degrees 45½ minutes East
N12.4530E   not  N12.455E
```

A fraction of a degree works the same way: half a degree is 30 minutes.

10. Sections first, then the tract

When a description is tied to a section, township, and range, draw the section first, then the tract inside it. Type the section and press Enter so the whole section appears, then add the tract.

```
all,9,16n,7w  draws the full section
```

This keeps the tract in context and lets you add more tracts to the same section later. If the map is only meant to show one tract, color that tract so it stands out: click inside it until it is outlined in green, then Fill.

11. Multiple tracts: chaining and merging

To show more than one tract, separate each tract's calls with `@0` on its own line. Type the first tract's calls, then `@0`, then the next tract's calls. That is chaining. Merging does the same automatically: use **File**, then **Merge**, to pull calls you saved in another file into the editor, and the program inserts the `@0` for you. Move or rotate tracts into position after they draw.

12. Turning deed call labels on and off

The **Show Deed Calls** icon toggles the call labels along the boundary lines. With no tract selected it toggles every tract. With a tract active it toggles only that one. To hide a single label that overlaps another, select the corner where the line starts, open the **Lines** menu, and toggle that line off.

13. Basis of bearing: a rotated section

Many sections, especially older ones, are rotated from true north, so the deed gives a basis of bearing. A standard section has due north, south, east, and west sides, and `all,9,16n,7w` draws it that way. When the deed gives a basis of bearing, match the tilt one of two ways.

Rotate the section. Build the standard section, then rotate it by the difference between due west and the basis of bearing, using the Rotate Selected Tract tool.

```
90°00'00"  (due west)
− 83°26'15"  (basis of bearing)
= 6°33'45"  rotate the section by this
```

Build it on the basis of bearing. Start with the deed's bearing instead of due west, turning each following side 90 degrees.

```
N83.2615W 5280
S6.3345W 5280
S83.2615E 5280
N6.3345E 5280
```

> Use the first basis of bearing in the deed, since it sets the section's overall orientation. A separate basis on an interior tract usually does not change a simple plot, and a tract sticking slightly past a section line is normal: the deed follows a real survey, while the section you draw is a theoretical PLSS square.